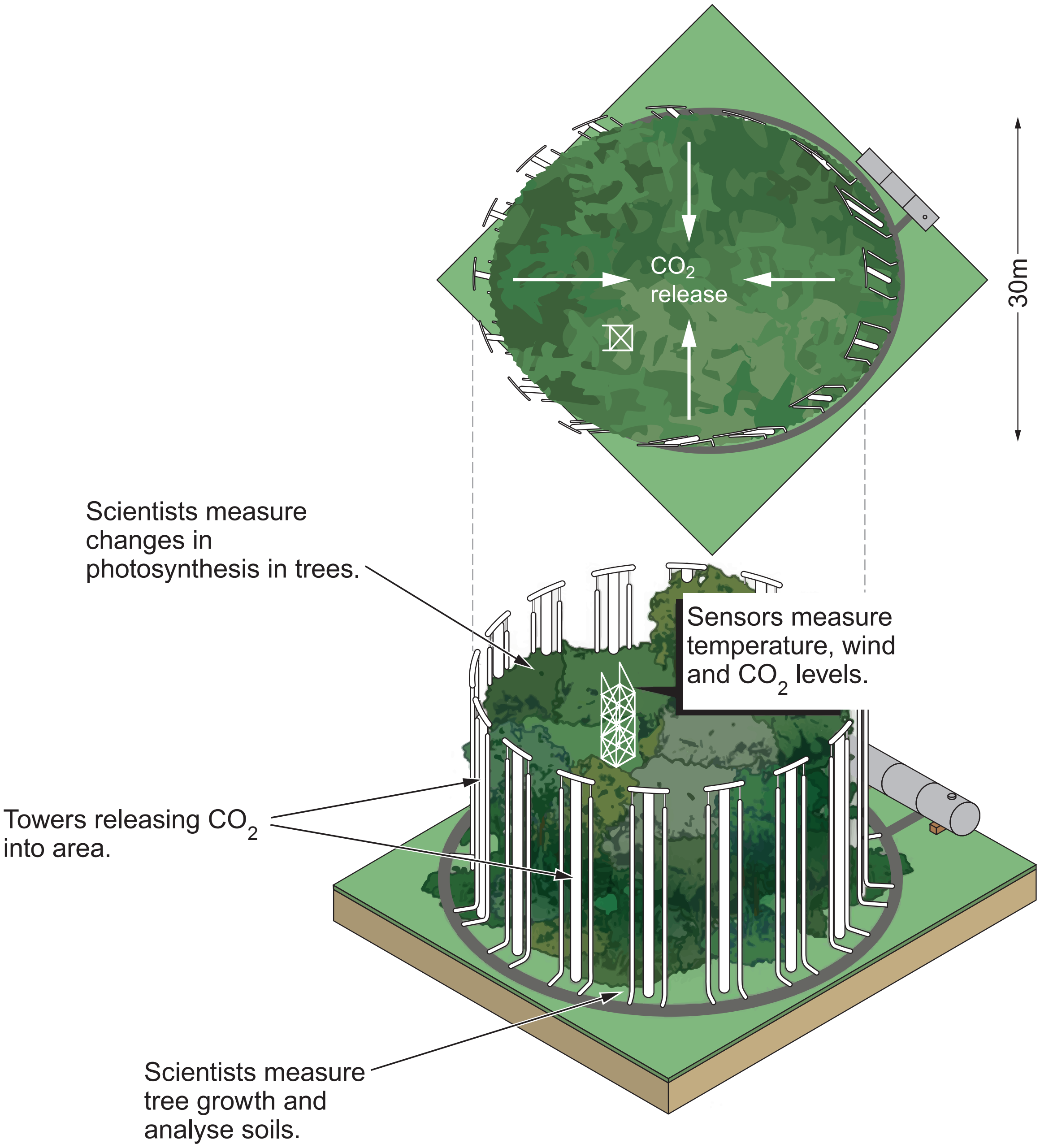
9. The concentration of carbon dioxide (CO_2) in the atmosphere has risen from a mean of 315 parts per million (ppm) in 1959 to a current mean of 385 ppm. It is predicted that the concentration of CO_2 will continue to rise to between 500 and 1000 ppm by the year 2100.

Most of our current knowledge on plant responses to CO_2 concentrations in the atmosphere is based on experiments carried out in greenhouses.

Free-air carbon dioxide enrichment (FACE) experiments have been carried out around the world to study the effects of increased CO_2 levels on photosynthesis in plants outside greenhouses. This will enable scientists to make conclusions about the effect of increased concentrations of CO_2 in the future.

Figure 1 – Illustration of a FACE experiment – aerial view (above), side view (below)



Scientists have used the results of many FACE experiments to predict the mass of CO_2 taken in by plants each year. **Graph 1** shows the effect of atmospheric CO_2 concentration on annual global uptake of carbon.

Graph 1